# Lecture 2: Sections 0.2 and 0.3

## Family of functions

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# Composition of Functions

## Definition

Given two functions $f$ and $g$, the composition $f \circ g$ is defined as:

$$(f \circ g)(x) = f(g(x))$$

## Example

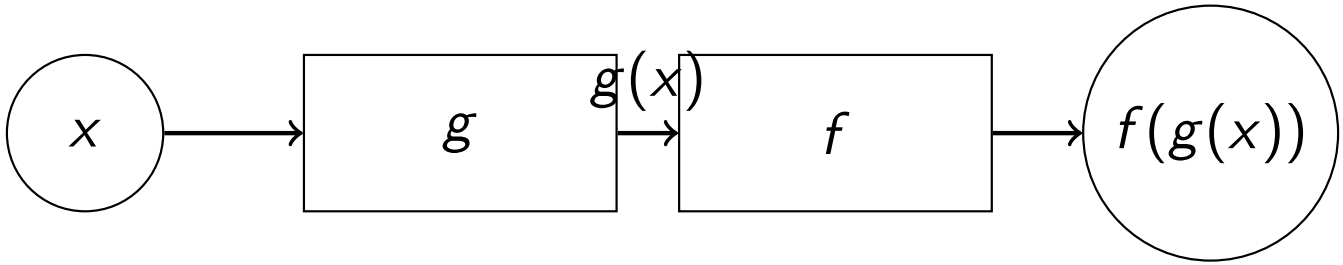
Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:

- $(f \circ g)(x)$
- $(g \circ f)(x)$
- $(f \circ f)(x)$
- $(g \circ g)(x)$

# Visualizing Function Composition

## Important Note

Composition is not commutative: $f \circ g \neq g \circ f$




## Example

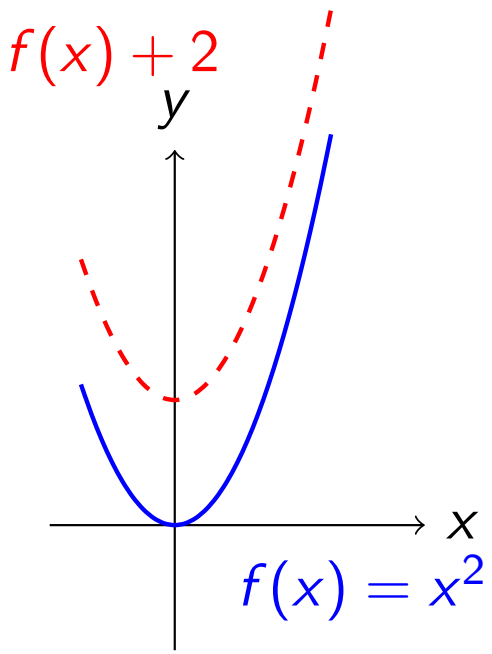
If $f(x) = \sqrt{x}$ and $g(x) = x^2 + 4$, find:

- Domain of $(f \circ g)(x)$
- Domain of $(g \circ f)(x)$

# Transformations: Translation

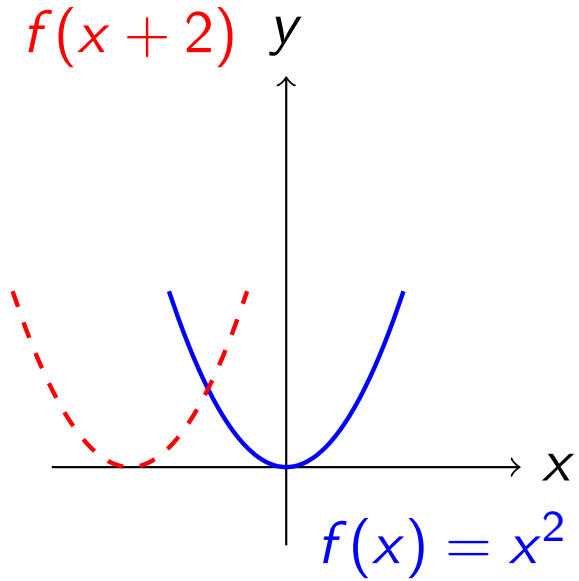
## Vertical Translation

$f(x) + c$: shift up by $c$ units
$f(x) - c$: shift down by $c$ units




## Horizontal Translation

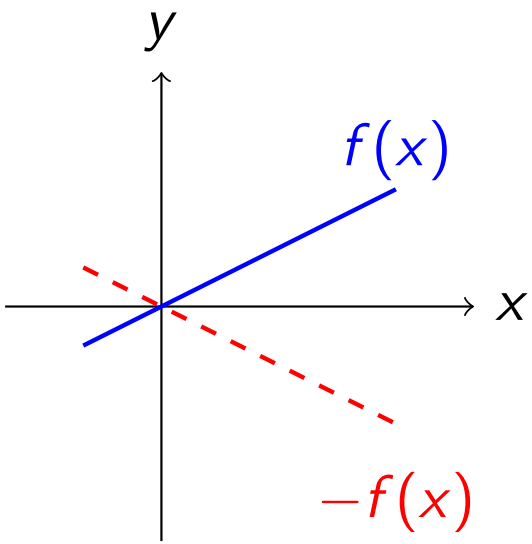
$f(x + c)$: shift left by $c$ units
$f(x - c)$: shift right by $c$ units



# Transformations: Reflection

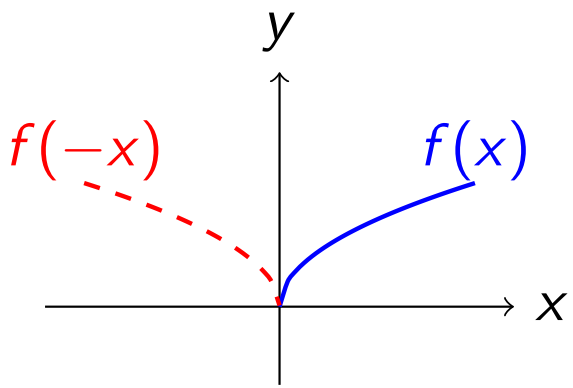
## Vertical Reflection

$-f(x)$: reflect over x-axis




## Horizontal Reflection

$f(-x)$: reflect over y-axis



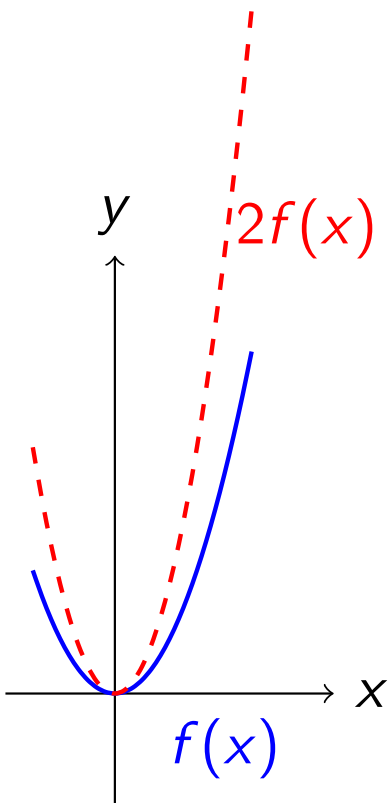
# Transformations: Stretches and Compressions

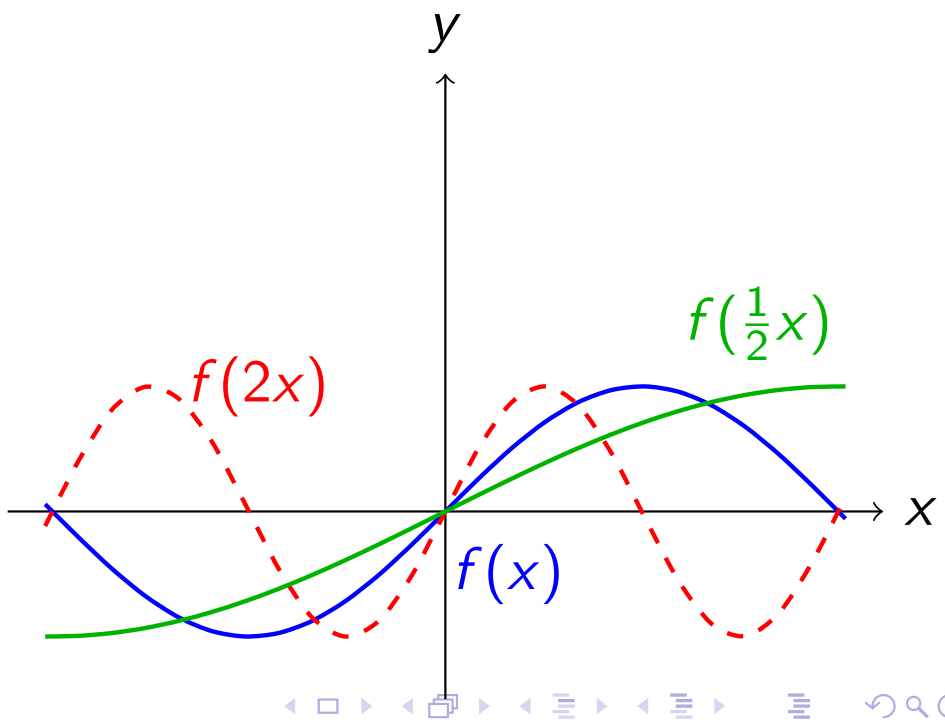
## Vertical Stretch/Compression

$cf(x)$:
$|c| > 1$: vertical stretch
$0 < |c| < 1$: vertical compression

## Horizontal Stretch/Compression

$f(cx)$:
$|c| > 1$: horizontal compression
$0 < |c| < 1$: horizontal stretch



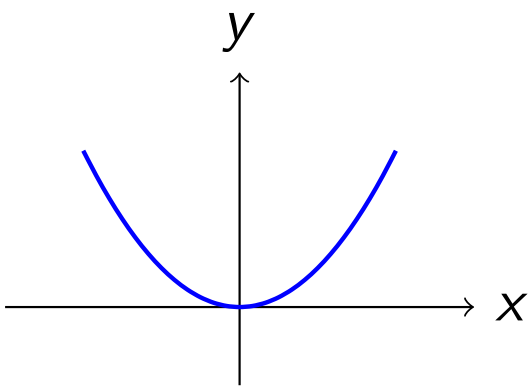



# Symmetry in Functions

## Even Functions

$f(-x) = f(x)$

Symmetric about y-axis

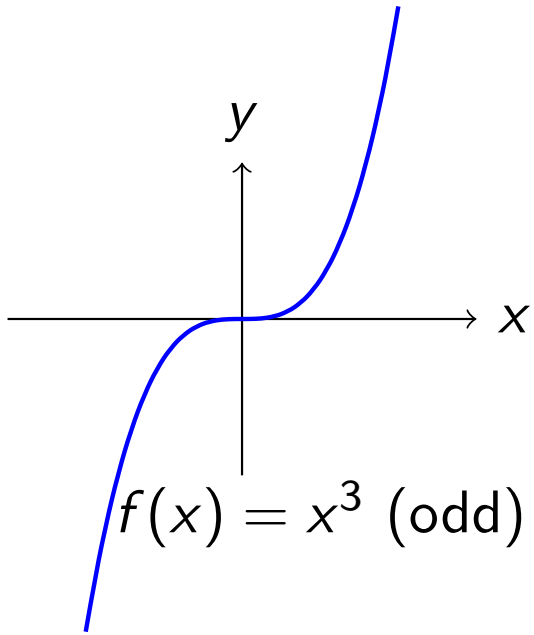



$f(x) = x^2$ (even)

## Odd Functions

$f(-x) = -f(x)$

Symmetric about origin




$f(x) = x^3$ (odd)

# Even and Odd Functions

## Definitions

- **Even Function**: $f(-x) = f(x)$ for all $x$ in domain
- **Odd Function**: $f(-x) = -f(x)$ for all $x$ in domain
- A function can be neither even nor odd

## Example

Determine if each function is even, odd, or neither:

- $f(x) = x^4 - 2x^2 + 1$
- $g(x) = x^3 - x$
- $h(x) = x^2 + x$
- $k(x) = \sin(x)$

# Practice Problems

## Composition and Transformations

1. Given $f(x) = \sqrt{x}$ and $g(x) = x^2 + 1$, find $(f \circ g)(x)$ and $(g \circ f)(x)$
2. Describe the transformations: $y = 2f(x - 3) + 1$
3. Graph $y = -|x + 2|$ using transformations

## Symmetry

1. Determine if $f(x) = x^3 \cos(x)$ is even, odd, or neither
2. Show that $f(x) = \frac{x}{x^2+1}$ is odd

# Applications and Summary

## Real-world Applications

- Composition: Chain reactions, multi-step processes
- Transformations: Signal processing, image manipulation
- Symmetry: Physics (conservation laws), engineering design

## Key Concepts

- Composition: $f(g(x))$ applies functions from inside out
- Translations: Shift graphs horizontally/vertically
- Reflections: Mirror images across axes
- Stretches/Compressions: Scale graphs vertically/horizontally
- Even functions: y-axis symmetry
- Odd functions: origin symmetry

# Rational Functions

## Definition

A **rational function** is a function of the form:

$$f(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$ and $Q(x)$ are polynomials and $Q(x) \neq 0$.

## Key Features

- **Vertical asymptotes**: Occur where denominator $= 0$ (and numerator $\neq 0$)
- **Horizontal asymptotes**: Describe end behavior as $x \to \pm\infty$

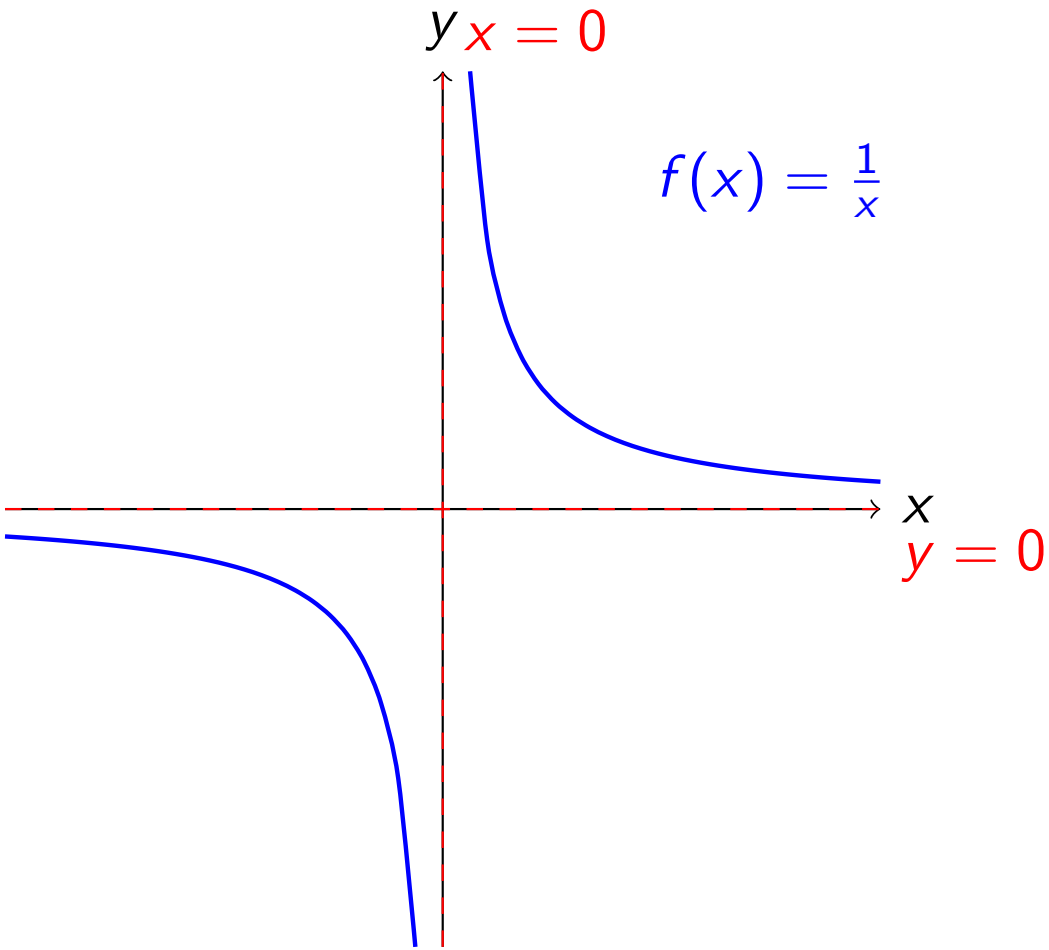
# Example

## Example

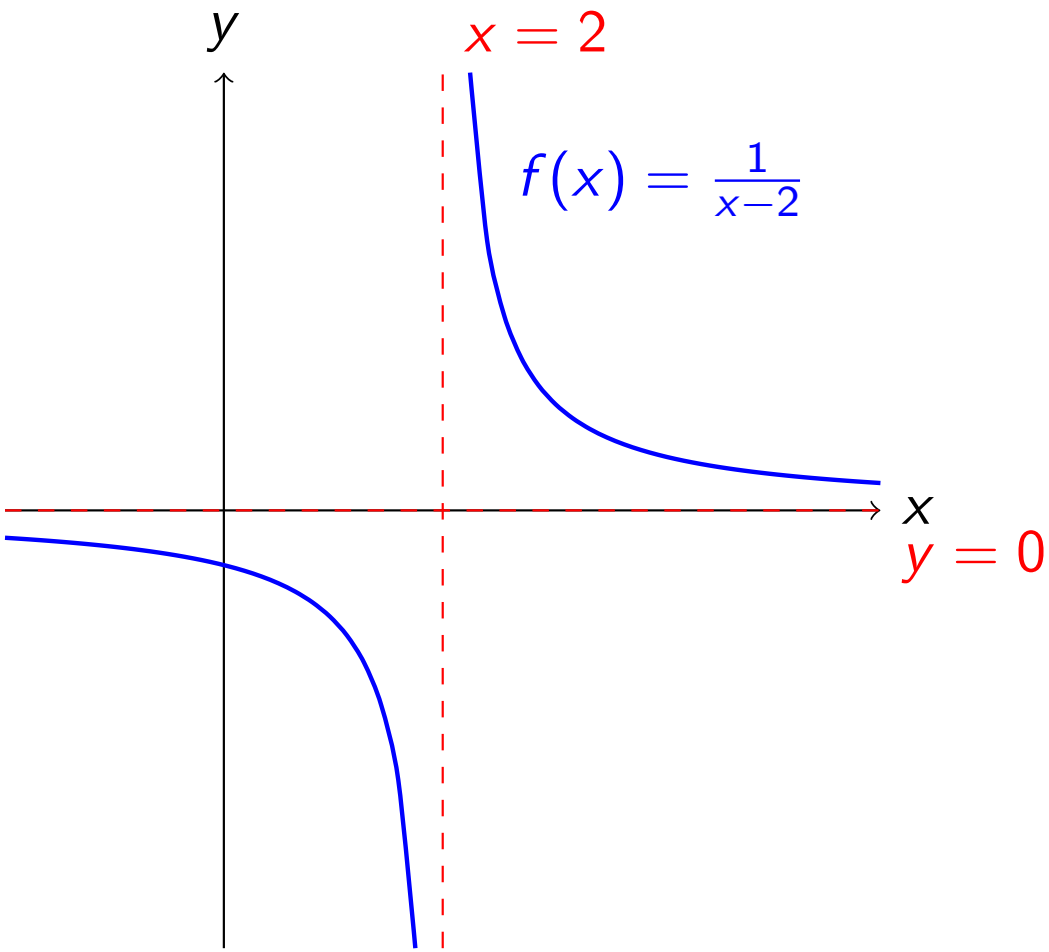
Graph the following rational functions:

1. $f(x) = \frac{1}{x}$
2. $f(x) = \frac{1}{x-2}$
3. $f(x) = \frac{x}{x^2-4}$
4. $f(x) = \frac{x^2-1}{x^2-4}$
5. $f(x) = \frac{2x+1}{x-3}$

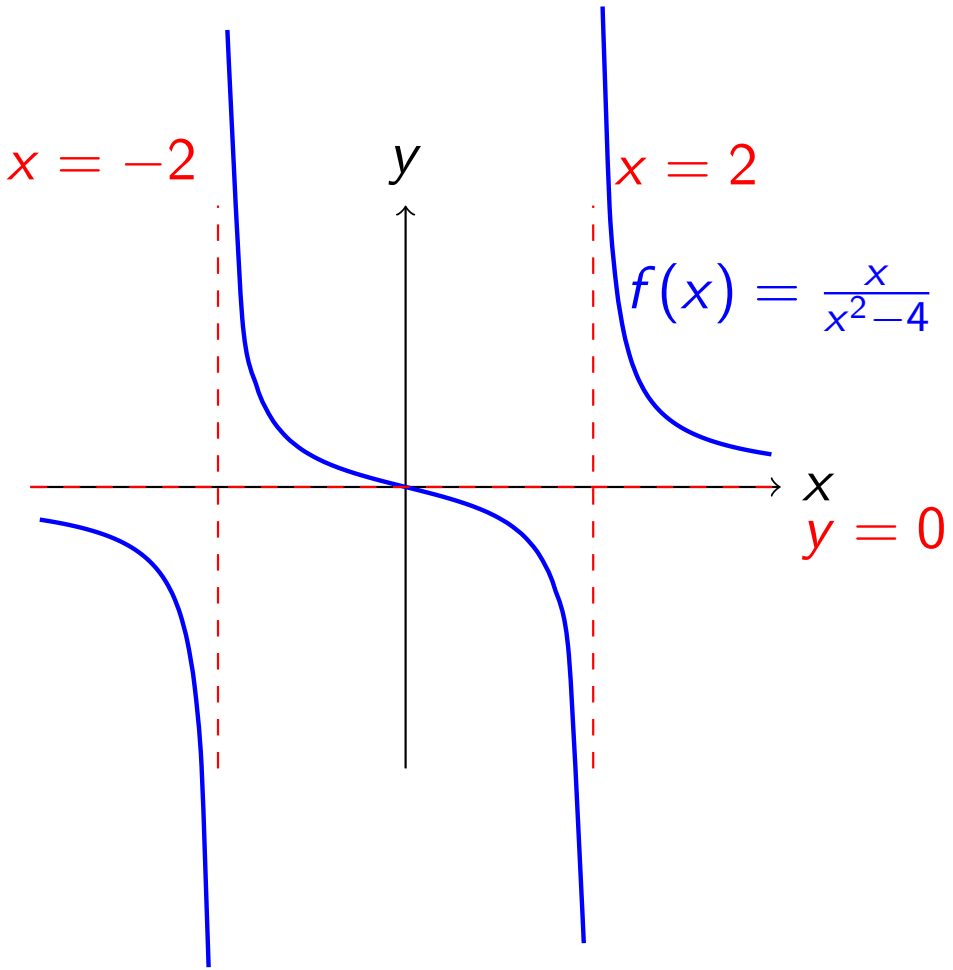
# Rational Function: $f(x) = \frac{1}{x}$



# Rational Function: $f(x) = \frac{1}{x-2}$



# Rational Function: $f(x) = \frac{x}{x^2-4}$



# Families of Trigonometric Functions

## General Forms

- $y = A\sin(Bx - C)$
- $y = A\cos(Bx - C)$

## Parameters

- $A$: **Amplitude** (vertical stretch/compression)
- $B$: Affects **period** and **frequency**
- $C$: **Phase shift** (horizontal translation)

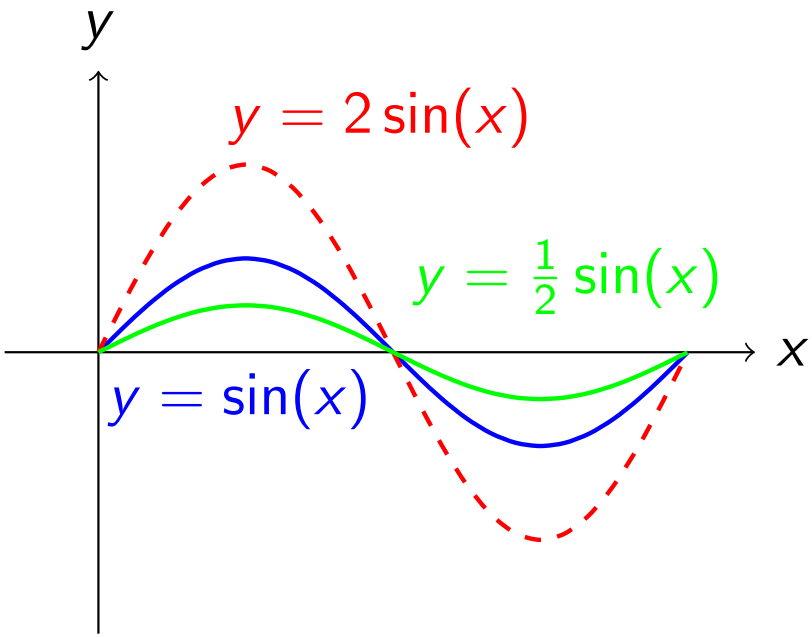
# Example

## Example

Graph the following trigonometric functions:

1. $y = 2\sin(x)$
2. $y = \sin(2x)$
3. $y = \cos(x - \frac{\pi}{2})$
4. $y = 3\cos(2x + \pi)$
5. $y = \frac{1}{2}\sin(4x - \pi)$

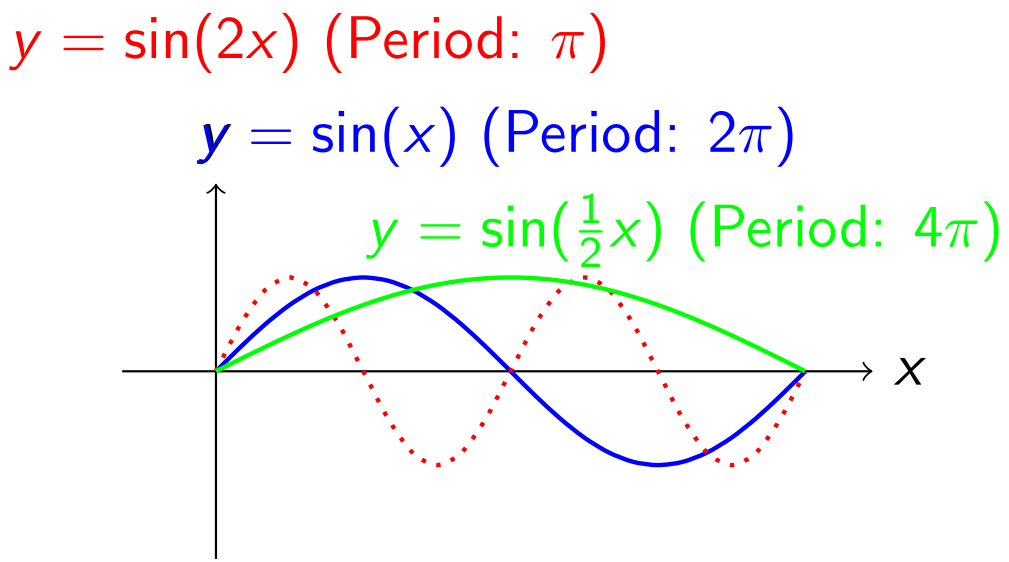
# Amplitude: $y = A\sin(x)$




## Amplitude Effect

- $|A|$ = maximum displacement from midline
- $A > 0$: normal orientation
- $A < 0$: reflected over x-axis

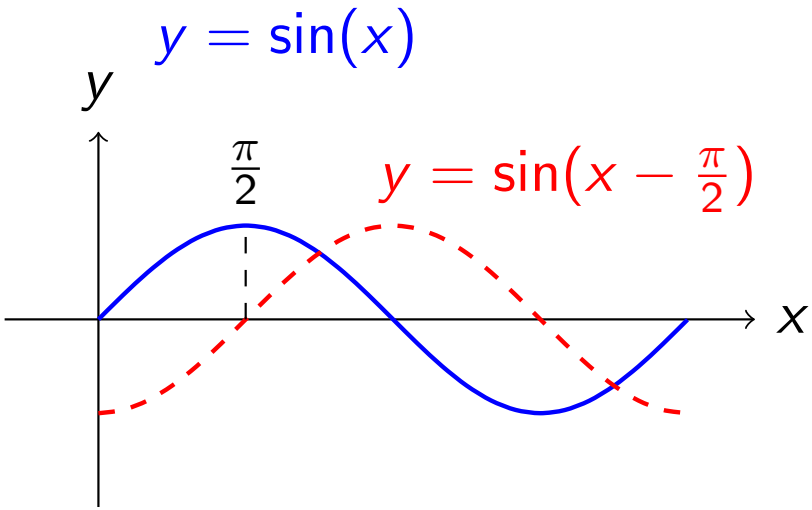
# Period and Frequency: $y = \sin(Bx)$




## Period and Frequency

- Period $= \frac{2\pi}{|B|}$
- Frequency $= \frac{|B|}{2\pi}$ (cycles per unit)
- $B > 1$: compressed horizontally (more cycles)
- $0 < B < 1$: stretched horizontally (fewer cycles)

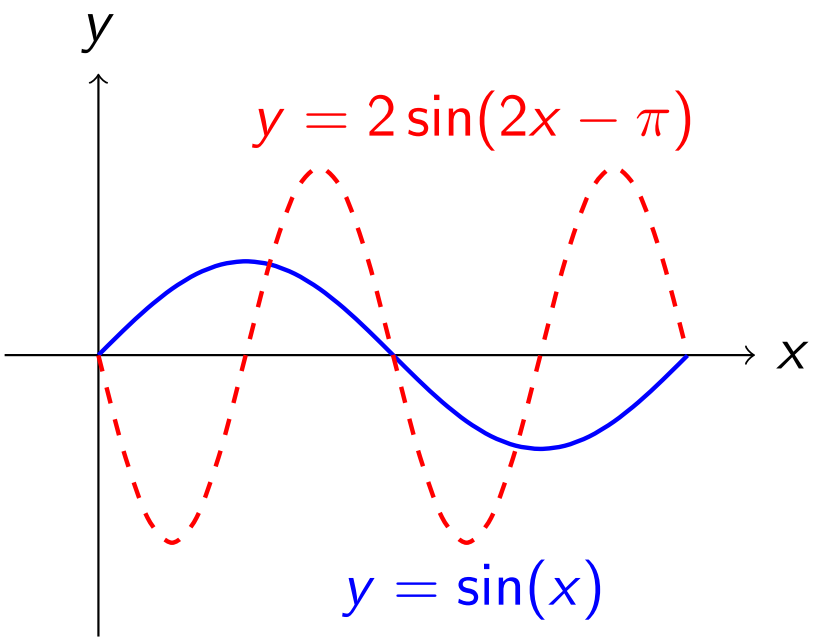
# Phase Shift: $y = \sin(x - C)$




## Phase Shift

- Phase shift $= \frac{C}{B}$
- $C > 0$: shift to the right
- $C < 0$: shift to the left
- For $y = \sin(Bx - C)$, rewrite as $y = \sin[B(x - \frac{C}{B})]$

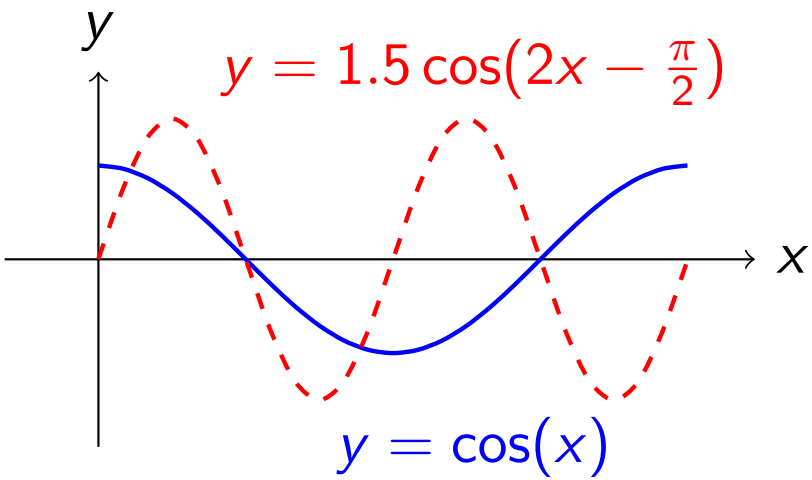
# Combined Transformations: $y = 2\sin(2x - \pi)$




## Analysis

- Amplitude: $|A| = 2$
- Period: $\frac{2\pi}{|B|} = \frac{2\pi}{2} = \pi$
- Phase shift: $\frac{C}{B} = \frac{\pi}{2}$ (right)
- Rewritten: $y = 2\sin[2(x - \frac{\pi}{2})]$

# Cosine Function: $y = A\cos(Bx - C)$




## Cosine vs Sine

- Same parameters (A, B, C) affect cosine similarly
- Cosine starts at maximum (when no phase shift)
- $\cos(x) = \sin(x + \frac{\pi}{2})$ (phase relationship)

# Practice Problems

## Rational Functions

1. Graph $f(x) = \frac{2x-1}{x+3}$ and identify asymptotes
2. Find domain of $g(x) = \frac{x^2-9}{x^2-4x+3}$

## Trigonometric Functions

1. Graph $y = 3\sin(\frac{1}{2}x + \pi)$
2. Find amplitude, period, and phase shift of $y = -2\cos(4x - \frac{\pi}{3})$
3. Write the equation for a sine function with amplitude 4, period $3\pi$, and phase shift $\frac{\pi}{2}$

# Applications

## Real-world Applications

- **Rational functions**: Engineering circuits, economics, physics
- **Trigonometric functions**: Sound waves, light waves, alternating current, seasonal patterns
- **Amplitude**: Volume (sound), brightness (light), voltage
- **Frequency**: Pitch (sound), color (light), radio stations
- **Phase shift**: Time delays, synchronization

## Key Formulas

- Amplitude $= |A|$
- Period $= \frac{2\pi}{|B|}$
- Frequency $= \frac{|B|}{2\pi}$
- Phase shift $= \frac{C}{B}$